

Notes: Gao, Sockin, & Xiong (RFS, 2021)

Learning about the Neighborhood

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1 Big picture

- **Question.** How do informational frictions in housing markets affect housing prices, migration into neighborhoods, construction, and local real investment?
- **Core idea.** Housing prices are not only prices of houses. They are also public signals about local economic strength. When people are unsure about a neighborhood, they learn from prices.
- **Model setting.** Households decide whether to move into a new neighborhood by buying a house at $t = 1$. The true neighborhood strength A is unknown at that time and is revealed only later.
- **Complementarity.** Households benefit from living near other productive households because they consume each other's locally produced goods and services. This makes each household care about who else moves in.
- **Key friction.** A high housing price may reflect strong neighborhood fundamentals, but it may also reflect housing-supply noise. When households and capital producers cannot perfectly separate these components, price learning can amplify noise.
- **Feedback loop.** A noisy high price makes households more optimistic, induces more entry, raises demand, supports higher prices, and encourages capital producers to invest more in local capital.
- **Supply elasticity prediction.** The distortion is strongest at intermediate housing-supply elasticity. If supply is fully inelastic, price reveals demand; if supply is fully elastic, price is pinned down by supply. The intermediate case creates the noisiest learning problem.
- **Complementarity prediction.** Stronger local consumption complementarity makes learning about the neighborhood more important, so the price boom-bust effect rises with complementarity.
- **Capital-investment prediction.** Stochastic learning affects not only housing but also local capital. Optimistic learning can generate overbuilding of both houses and commercial/local capital.
- **Empirical illustration.** Across U.S. MSAs during the 2000s housing cycle, price booms, price busts, and new construction are most pronounced in intermediate-supply-elasticity areas. Price cycles are stronger in high-complementarity areas.
- **Economic takeaway.** Housing-market noise can become real economic noise because households and firms use prices to learn, and their decisions are strategic complements.

2 Data, measurement, and model objects

2.1 Why housing markets are a natural information-aggregation setting

- A neighborhood's economic strength is hard to observe before moving in.
- A household observes its own opportunity in the neighborhood but does not fully know whether that opportunity comes from its own type or the neighborhood's common strength.
- The housing price aggregates information from demand and supply, but the two sides are mixed together.

- Unlike a standard asset market, the identity of owners matters: if more productive households move in, the neighborhood becomes more valuable for everyone else.

Interpretation. The housing price is a signal about fundamentals, but it is an imperfect signal. Because households' entry choices are complementary, a noisy signal can move real activity.

2.2 Agents and timing

- There are two periods, $t \in \{1, 2\}$, and one neighborhood.
- The economy has three types of agents: households, home builders, and capital producers.
- At $t = 1$, households decide whether to enter the neighborhood and buy one house at price P .
- At $t = 1$, home builders decide whether to build houses, and capital producers choose how much local capital to develop.
- At $t = 2$, the neighborhood strength is revealed; households produce goods, trade with one another, supply labor, and use local capital.

Interpretation. The model is static in the sense of having one move-in decision and one production period, but it is designed to capture a long episode in which a generation of households enters a neighborhood and then interacts locally.

2.3 Households, productivity, and complementarity

- Household i can live inside or outside the neighborhood. The outside option is normalized to zero.
- If household i enters, $H_i = 1$; otherwise $H_i = 0$.
- Household productivity is $A_i = A + \varepsilon_i$, where A is common neighborhood strength and ε_i is idiosyncratic productivity.
- The common component A is the demand-side fundamental of the neighborhood.
- The parameter $\eta_c \in (0, 1)$ measures the weight on consuming goods made by other households; higher η_c means stronger complementarity.

Interpretation. Complementarity is the source of strategic interaction. A household wants to know not only its own productivity but also the productivity of neighbors who will provide local services and trading opportunities.

2.4 Home builders and supply elasticity

- Home builders differ in construction productivity.
- The supply side contains a common shock $\xi = k\zeta$ and idiosyncratic builder shocks.
- The parameter $k > 0$ governs housing-supply elasticity.
- A higher k makes supply more elastic and makes the housing price more affected by supply-side forces.

Interpretation. Supply elasticity matters twice: it changes physical construction responses and it changes the information content of the price. The latter role is the paper’s distinctive mechanism.

2.5 Capital producers

- Capital producers develop local capital such as offices, warehouses, equipment, and infrastructure.
- They choose the amount of capital at $t = 1$, before the neighborhood strength A is fully known.
- The realized capital price at $t = 2$ depends on the marginal product of capital across households in the neighborhood.
- Therefore, capital producers also learn from the housing price.

Interpretation. Capital makes the housing signal affect the broader local economy. A price-driven belief about neighborhood strength can generate too much local capital and commercial real estate.

2.6 Empirical measurement

- The empirical section uses the recent U.S. housing cycle: boom period 2001–2006 and bust period 2007–2010.
- MSA-level house prices come from the Federal Housing Finance Agency repeat-sales House Price Index.
- Housing-supply elasticity comes from Saiz’s geography-based measure, available for 269 MSAs.
- New housing supply is measured by building permits from 2001 to 2006 divided by existing housing units in 2000.
- Complementarity is proxied by the MSA employment share in nontradable industries, constructed from County Business Patterns data and following Mian and Sufi’s nontradable-industry classification.
- The paper separately flags “sand states” - Arizona, California, Florida, and Nevada - because they experienced unusually strong speculative housing cycles.

Interpretation. The empirical exercise is designed as stylized evidence for the model, not as a full causal test. The key object is the cross-sectional shape of boom, bust, and construction outcomes across MSAs.

3 Models and empirical specifications

3.1 Household utility

Household i consumes its own good $C_i(i)$ and goods produced by other households in the neighborhood:

$$U(\{C_j(i)\}_{j \in N}; N) = \left(\frac{C_i(i)}{1 - \eta_c} \right)^{1 - \eta_c} \left(\frac{\int_{N/i} C_j(i) dj}{\eta_c} \right)^{\eta_c}. \quad (1)$$

Interpretation. The utility function makes local composition valuable. If other households are more productive, their goods and services are more attractive, so the neighborhood becomes more valuable for household i .

3.2 Household production

Household i produces with local capital and labor:

$$Y_i = e^{A_i} K_i^\alpha l_i^{1-\alpha}, \quad A_i = A + \varepsilon_i. \quad (2)$$

Interpretation. The common component A shifts productivity for all households. The idiosyncratic component ε_i determines how attractive the neighborhood is for a particular household.

3.3 Information and entry

At $t = 1$, households observe a private signal embedded in A_i , the public signal Q , and the housing price P :

$$Q = A + \tau_Q^{-1/2} \varepsilon_Q, \quad I_i = \{A_i, P, Q\}. \quad (3)$$

The entry decision is

$$H_i = \mathbf{1} \{E[U_i | I_i] - P \geq 0\}. \quad (4)$$

Interpretation. The house price is both a cost of entering and an information signal. A high price discourages entry through the cost channel but encourages entry through the belief channel when it is interpreted as evidence of high A .

3.4 Capital producers

The representative capital producer chooses capital at $t = 1$:

$$\Pi_c = \max_K E \left[RK - \frac{1}{\lambda} K^\lambda \mid P, Q \right], \quad K = E[R \mid P, Q]^{1/(\lambda-1)}. \quad (5)$$

Interpretation. Capital supply increases when the producer's posterior belief about neighborhood strength rises. Housing-price noise therefore affects the stock of local capital.

3.5 Home builders

Builder i supplies a house when the price covers its cost:

$$S_i = \mathbf{1} \left\{ P \geq \exp \left[-\frac{k\zeta + e_i}{1+k} \right] \right\}. \quad (6)$$

Interpretation. The common building-cost shock is a source of supply-side noise. Households observe the final price, but not the exact decomposition between demand strength and supply conditions.

3.6 Cutoff equilibrium

The equilibrium is a cutoff equilibrium: households enter if their productivity exceeds a threshold,

$$H_i = \mathbf{1}\{A_i \geq A^*\}. \quad (7)$$

Market clearing between household demand and builder supply gives the housing-price equation

$$\log P = \frac{1}{1+k} \left[\sqrt{\frac{\tau_\varepsilon}{\tau_e}} (A - A^*) - \xi \right]. \quad (8)$$

Interpretation. The price rises with neighborhood strength A , falls with the cutoff A^* because fewer households enter, and falls with the adverse supply shock ξ .

3.7 Price as a noisy sufficient statistic

The public information in the housing price can be summarized by

$$z(P) = A - \sqrt{\frac{\tau_e}{\tau_\varepsilon}} \xi = (1+k) \sqrt{\frac{\tau_e}{\tau_\varepsilon}} \log P + A^*. \quad (9)$$

Interpretation. The price reveals a mixture of demand strength and supply shocks. The signal is especially noisy when demand and supply both matter substantially for the price.

3.8 Posterior learning

Given P , Q , and its private signal A_i , household i forms a Gaussian posterior about A :

$$\hat{A}_i = \frac{\tau_A \bar{A} + \tau_Q Q + (\tau_\varepsilon/\tau_e) \tau_\xi z + \tau_\varepsilon A_i}{\tau_A + \tau_Q + (\tau_\varepsilon/\tau_e) \tau_\xi + \tau_\varepsilon}. \quad (10)$$

Capital producers do not observe household i 's private signal, so their posterior mean is

$$\hat{A}_c = \frac{\tau_A \bar{A} + \tau_Q Q + (\tau_\varepsilon/\tau_e) \tau_\xi z}{\tau_A + \tau_Q + (\tau_\varepsilon/\tau_e) \tau_\xi}. \quad (11)$$

Interpretation. Households and capital producers both learn from the price statistic z , but households also have private information through their own productivity draw.

3.9 Learning amplification

A public-signal shock affects the price through the equilibrium cutoff:

$$\frac{\partial \log P}{\partial Q} = -\frac{1}{1+k} \sqrt{\frac{\tau_\varepsilon}{\tau_e}} \frac{\partial A^*}{\partial Q}, \quad \frac{\partial A^*}{\partial Q} < 0. \quad (12)$$

A shock to the price statistic z affects the price as

$$\frac{\partial \log P}{\partial z} = \frac{1}{1+k} \sqrt{\frac{\tau_\varepsilon}{\tau_e}} \left(1 - \frac{\partial A^*}{\partial z} \right). \quad (13)$$

Interpretation. If better signals lower the entry cutoff, more households enter. That entry response can amplify the original signal in the housing price.

3.10 Core comparative statics

- A positive noise shock to public information raises beliefs, lowers the cutoff, raises housing prices, and raises housing stock.
- The price distortion from noise is hump shaped in supply elasticity.
- A negative supply shock can be misread as a positive demand signal, again raising beliefs and amplifying price movements.
- Higher complementarity increases the importance of learning about A , so price responses become larger as η_c rises.
- When capital investment is included, mislearning in the housing market also distorts capital supply and reinforces the housing response.

Interpretation. The model generates boom-like and bust-like distortions without assuming mechanically extrapolative expectations. The extrapolation-looking behavior comes from rational learning under noisy signals and complementarities.

4 Identification and empirical design

4.1 What the empirical exercise identifies

- The paper does not claim a sharp causal estimate of learning from housing prices.
- Instead, it asks whether the cross-section of the 2000s U.S. housing cycle matches distinctive model predictions.
- The key empirical objects are the shapes of boom, bust, and construction outcomes across supply elasticity and complementarity.

Interpretation. The empirical exercise is a model-validation exercise: do the puzzling cross-sectional patterns look like the model's predictions?

4.2 Why supply elasticity is useful

- Saiz's supply-elasticity measure is based on geographical constraints such as steep terrain and bodies of water.
- This makes it less directly tied to contemporaneous demand shocks than realized housing construction.
- The model predicts a nonmonotonic relation: the strongest price cycles and construction should occur in intermediate-elasticity MSAs.

Interpretation. A monotonic supply story would predict very constrained areas have the largest price booms and very elastic areas have the most construction. The paper instead emphasizes the intermediate range.

4.3 Why complementarity is useful

- The nontradable employment share proxies for the importance of local interactions and local demand.
- More nontradables mean households rely more on local goods and services produced by other residents.
- The model predicts stronger price-cycle magnitudes with complementarity, and a hump-shaped relation for construction.

Interpretation. Complementarity is hard to measure directly. Nontradable employment is a practical proxy for the local-service intensity of an MSA.

4.4 Sand-state robustness

- Arizona, California, Florida, and Nevada experienced unusually large housing cycles and speculation.
- The figures mark sand-state MSAs separately and show separate spline lines excluding them.
- The main nonmonotonic patterns remain, although with attenuated magnitudes.

Interpretation. The results are not only a Las Vegas/Phoenix-type story. Sand states matter, but the intermediate-elasticity pattern persists outside them.

4.5 What the paper cannot fully prove

- The model is static and does not literally generate a multi-period boom and bust.
- The empirical section does not isolate exogenous shocks to beliefs or learning.
- Other forces in the 2000s housing cycle, including credit expansion, securitization, foreign capital inflows, and low interest rates, also mattered.
- Complementarity is proxied rather than directly observed.
- The evidence supports the mechanism but does not uniquely rule out all alternative explanations.

Interpretation. The paper's contribution is a new mechanism plus cross-sectional evidence that lines up with it. The empirical facts are best read as suggestive and disciplined by theory.

5 Figures

5.1 Figures 1 and 2: Cutoff responses and public-signal noise

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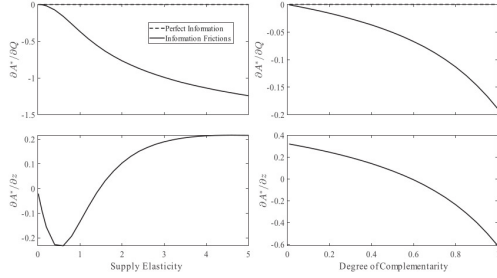


Figure 1
Response of the equilibrium cutoff productivity to a noise shock Q (the first row) and a fundamental shock z (the second row) across housing supply elasticity (left) and degree of complementarity (right). The dotted line in each panel represents the perfect-information benchmark, and the solid line represents the case with informational frictions.

Learning about the Neighborhood

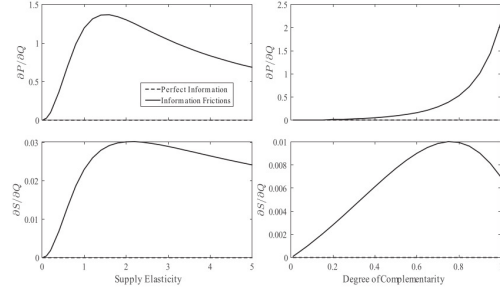


Figure 2
The responses of housing price P (top row) and housing stock S (bottom row) to a noise shock to the public signal Q across supply elasticity (left) and degree of complementarity (right). The dotted line in each panel represents the perfect-information benchmark, and the solid line represents the case with informational frictions.

Figure 1: Original Figures 1 and 2 from Gao, Sockin, and Xiong, cropped from the paper and grouped to save space.

Read:

- Figure 1 shows how the equilibrium cutoff A^* responds to noise in Q and to the fundamental price statistic z .
- In the perfect-information benchmark, the public noise shock has no effect.
- With information frictions, positive noise lowers the cutoff, especially when supply elasticity or complementarity is high.
- The response to z is most negative at intermediate supply elasticities, which means a better price signal induces more entry and amplifies the price.
- Figure 2 shows that a noise shock raises both housing prices and housing stock.
- The price and stock responses are hump shaped in supply elasticity and stronger with complementarity.

Economic message. The first pair of figures establishes the main learning mechanism: noisy optimism enters through beliefs, changes the entry cutoff, and produces real housing-market responses.

5.2 Figures 3 and 4: Supply shocks and capital amplification

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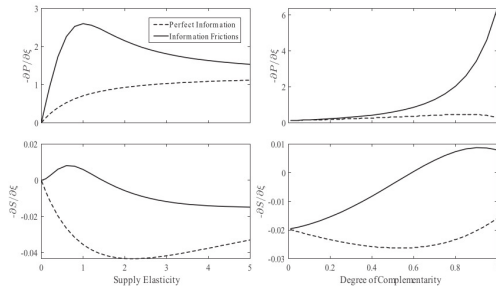


Figure 3
Responses of housing price, P (top row), and housing stock, S (bottom row), to a negative supply shock across supply elasticity (left) and degree of complementarity (right). The dotted line in each panel represents the perfect-information benchmark, and the solid line represents the case with informational frictions.

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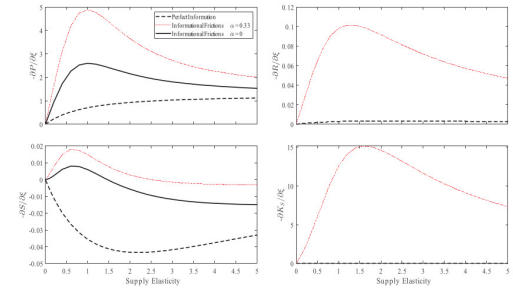


Figure 4
Responses of housing price, P (upper-left panel), housing stock, S (lower-left panel), capital price R_1 (upper-right panel), and capital stock, K (lower-right panel), to a negative housing supply shock ξ . In each panel, the dashed line represents the perfect-information benchmark; the solid line for the case with informational frictions and $\alpha=0.33$; and the dotted line represents case with informational frictions and $\alpha=0$.

Figure 2: Original Figures 3 and 4 from Gao, Sockin, and Xiong, cropped from the paper and grouped to save space.

Read:

- Figure 3 studies a negative housing-supply shock.
- Under perfect information, the shock raises housing prices and lowers housing stock in the expected way.
- With information frictions, households partially interpret the higher price as good news about demand, which amplifies the price boom and can raise construction relative to the benchmark.
- The distortion is again strongest at intermediate supply elasticity.
- Figure 4 adds capital investment with $\alpha = 0.33$ and compares it with the no-capital case.
- Capital producers also learn from the housing price, so they overinvest when the price signal is too optimistic.
- Capital overhang makes housing demand more aggressive because households anticipate cheaper or more abundant capital in the future.

Economic message. The mechanism is not only about house prices. A noisy price signal can propagate into construction and local investment, generating synchronized housing and commercial-real-estate cycles.

5.3 Figures 5 and 6: U.S. housing-cycle evidence

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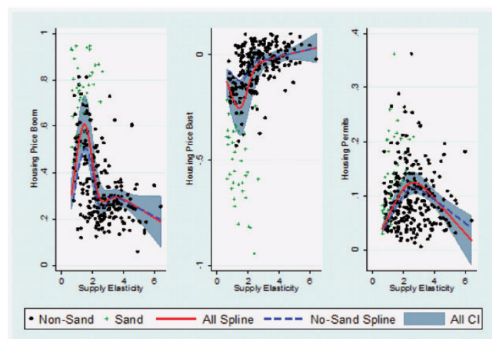


Figure 5
U.S. housing cycle in the 2000s across MSAs with different supply elasticities
The solid and cross dots represent MSAs outside and inside sand states (Arizona, California, Florida, and Nevada), respectively. The solid line represents the spline line for all MSAs, and the dashed line represents MSAs outside sand states. Ninety-five percent confidence intervals are displayed for the full sample. The standard errors are clustered at the state level.

Learning about the Neighborhood

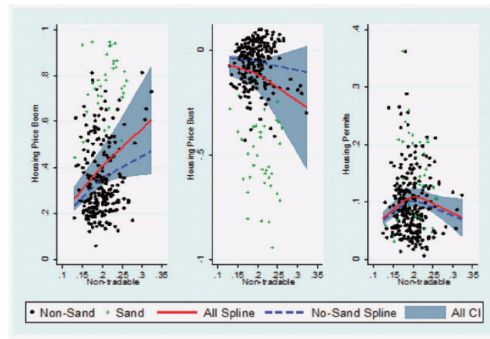


Figure 6
U.S. housing cycle in the 2000s across MSAs with different degrees of complementarity
The solid and cross dots represent MSAs outside and inside sand states (Arizona, California, Florida, and Nevada), respectively. The solid line represents the spline line for all MSAs, and the dashed line represents the spline line for MSAs outside sand states. Ninety-five percent confidence intervals are displayed for the full sample. The standard errors are clustered at the state level.

Figure 3: Original Figures 5 and 6 from Gao, Sockin, and Xiong, cropped from the paper and grouped to save space.

Read:

- Figure 5 plots the MSA-level housing price boom, housing price bust, and housing permits against housing-supply elasticity.
- The boom is hump shaped in supply elasticity, and the bust is U-shaped: intermediate-elasticity MSAs have the largest cycle.
- New housing permits are also hump shaped in supply elasticity, rather than simply increasing with elasticity.
- The sand-state observations are extreme, but excluding them leaves the key shape intact.
- Figure 6 plots the same outcomes against complementarity proxied by nontradable employment share.
- Price-cycle magnitudes increase with complementarity, while housing permits have a hump-shaped pattern.

Economic message. These figures are the empirical anchor of the paper. The cross-sectional shapes are difficult for a simple supply-constraint story but are natural in a model where price learning is noisiest at intermediate supply elasticity and strongest when local complementarities are high.

6 Short summary

- The paper builds a noisy rational expectations model of housing-market learning.
- Households are uncertain about neighborhood strength and learn from housing prices.
- Housing prices mix demand fundamentals with supply-side noise, so learning can be distorted.

- Household entry is complementary because residents consume one another's local goods and services.
- Noise-induced optimism can lower the entry cutoff, raise prices, increase housing stock, and encourage local capital production.
- The price distortion is hump shaped in housing-supply elasticity: intermediate-supply areas are the most vulnerable to noisy learning.
- Complementarity strengthens the learning feedback, producing larger price booms and busts.
- The empirical section shows that the 2000s U.S. housing cycle has the predicted nonmonotonic pattern across MSA supply elasticity and the predicted stronger price-cycle pattern across complementarity.

7 Conclusion

7.1 Core conclusion

Housing prices can transmit and amplify noise because agents use them to learn about local fundamentals. When entry and investment decisions are complementary, a noisy housing-market signal can move real migration, construction, and local capital investment.

7.2 Economic intuition

A high price is ambiguous. It could mean the neighborhood is strong, or it could mean supply is tight or noisy. When households interpret the price as good news, more of them enter. Their entry makes the neighborhood more valuable, and capital producers respond by investing. This endogenous response makes the initial price signal more powerful than it would be in a standard market.

7.3 Takeaway

The paper gives a theory of how housing booms can be both financial and real: expectations formed from housing prices affect who moves in, how much housing gets built, and how much local capital is supplied. The most fragile places are not necessarily the most supply constrained or the most supply elastic, but those in the intermediate region where prices are informative enough to be used and noisy enough to mislead.